MELANOMA SKIN CANCER DETECTION SYSTEM USING DEEP LEARNING

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By

Asad Ali	Registration No: 20-AU-LKM-234
Zeeshan Hassan	Registration No: 20-AU-LKM-232
Maheleqa Fazal	Registration No: 20-AU-LKM-240

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Session (2020-2024)

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Zeeshan Hassan	Registration No: 20-AU-LKM-232
Maheleqa Fazal	Registration No: 20-AU-LKM-240

Presented to Abdul Wali Khan University in partial fulfillment of the academic Requirements for obtaining the degree of

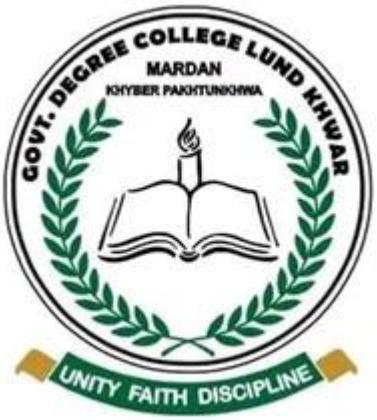
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FINAL APPROVAL

Our recommendation is to approve the project "Melanoma Skin Cancer Detection System Using Deep learning" that was presented by Mr. Asad Ali, Mr. Zeeshan Hassan and Ms. Maheleqa Fazal in order to fulfill the requirements for the degree of BS (CS).

Examination Committee: Signature: _____

Supervisor:
Mr. Mutabar Khan Signature: _____

Chairman Department of Computer Science: Signature: _____

CERTIFICATE

We declare that this thesis is the result of our own original work and has not been copied from any source, either in whole or in part. We have developed this project and the accompanying report entirely through our personal efforts, under the dedicated guidance of our supervisors. We confirm that no part of this work has been submitted for any other degree or qualification at this or any other university or institute. If it is discovered otherwise, we accept full responsibility.

Signature: _____
Name: Asad Ali
Registration No: 20-AU-LKM-234

Signature: _____
Name: Zeeshan Hassan
Registration No: 20-AU-LKM-232

Signature: _____
Name: Maheleqa Fazal
Registration No: 20-AU-LKM-240

DEDICATION

We dedicate this effort to our parents, who support us so much and hope for us to succeed in all that we do. We could never have been who we are today without the support, love, and right direction of our parents, with faith the All-Powerful Allah.

Currently, we are incredibly appreciative of all of our teachers, but especially to Mr. Mutabar Khan Chairman Department of computer science, whose instructions, mentoring, and guidance helped us to work in this innovative field of AI.

We are incredibly grateful to all of our family members and friends who helped and encouraged us with their wise counsel.

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First of all we are very thankful to Almighty Allah, The most beneficent and the king of all world and all mankind for his endless rewards, brainy capability and energy to accomplish this project. Countless blessings upon Prophet Muhammad (PBUH) the last Prophet of Allah, Rahmat ul lil Alameen, the mediator of Ummah. Using this amazing opportunity, we would like to express our gratitude to our supervisor Mr. Mutabar Khan Chairman Department Of Computer Science GDC Lundkhwar, for all his help, advice, tolerance, and constant support. His steadfast belief in us was a genuine source of inspiration, we would like to thank each and every one of our instructors for helping us in this journey and sharing their invaluable insights with us so that we could further develop our concept. We would like to thank our parents for their unwavering love, care and support all of which helped us to complete this project. We will always be grateful to them for everything that they have done for us.

Asad Ali

Zeeshan Hassan

Maheleqa Fazal

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Abbreviations

CNN	Convolutional Neural Network
MDS	Melanoma Detection System
MSC	Melanoma Skin Cancer
DL	Deep Learning
SDLC	System Development Lifecycle

ABSTRACT

Melanoma is highly aggressive variant of skin cancer, and early detection is essential for patient treatment outcomes. This project emphasizes the use of artificial intelligence, particularly DL for the timely identification of MSC. In advancement of technology in healthcare, AI-based detective models have shown outstanding promise in different medical fields. In this project some Deep Learning algorithms were used for identification of MSC. This project also include a user-friendly interface, developed using Django and MySQL, it will help both healthcare professionals and individuals to easily upload images and will quickly receive predictions. Among the algorithms tested, DenseNet-121 achieved the highest performance with an accuracy of 93.04%. The conclusion of this project specify that DL models can play a decisive role in early detection of melanoma, helping to save lives by ensuring quicker diagnosis.

Keywords: melanoma skin cancer, artificial intelligence, deep learning, DenseNet-121, Django.

CHAPTER 1
INTORDUCTION

1.1 Introduction

Melanoma is one of the most deadly type of skin cancer, and the chance of a successful treatment plan increases significantly with early identification. Traditional diagnostic approaches for dermatologists involve manual examinations, which can be time-consuming and prone to human mistake. The potential for automated detection systems to support early diagnosis has attracted a lot of attention because DL and artificial intelligence are developing at a rapid pace. Using DL techniques, this research aims to create a system that can identify MSC. The use of models like DenseNet-121 will improve detection process. We developed an easy-to-use web interface to let researchers and healthcare professionals use this technology even more.

1.2 Problem Statement

Melanoma can be challenging to identify, especially in its early stages, because it can outwardly mimic benign skin lesions. Due to dermatologists' limited availability and the human error risk associated with manual detection, early diagnosis is difficult in many places. The primary problem is the absence of automated, trustworthy, and easily navigable technologies that may aid healthcare providers in accurately and rapidly diagnosing melanoma, hence reducing false positive rates and ensuring timely treatment.

1.3 Aim and Objectives

Aim of this project is to develop a detection system through that medical professionals can detect melanoma in seconds and help them in early diagnoses of melanoma patients.

The project's main objectives are as follows:

- To develop a DL MDS capable of accurately identifying skin cancer.
- To implement DenseNet-121 as the primary model to reliably and accurately classify melanoma.
- To create an intuitive web-based application for users to facilitate them in early detection of melanoma.

1.4 Scope of the Project

The major objective of this study is to design and create an AI-powered MDS. The key features of the system are below:

- Model performance is improved by preprocessing the images.
- Melanoma and non-melanoma skin images are used for training and testing the DL models, especially a pre-trained model DesnseNet-121.
- A user-friendly web interface for uploading images, preprocessing and showing results of detected images which may be melanoma and non-melanoma with the predicted value between 0 and 1.
- MySQL database for storing the patient record with images when uploads linked with web interface.
- The system does not show or give any medical advice it is just a tool for detection of MSC it just shows the result of the uploaded image and its predicted value.

1.5 Significance of the Project

Struggling for and doing effort for changing the way of diagnosing the melanoma through a system makes it significant. By using DL algorithms the system may provide a high degree of accuracy, it reduces the need for manual checking and provide facility of early detection which is a great benefit of this system. This system provides amazing facilities in places where reaching of expert dermatologists is not possible and limited. Because of user-friendly interface of the system, Medical professionals can be easily trained to operate this system for detection of MSC. And further this system is integrated with MySQL database for storing patient details in which after we can easily checks patient record such as patient id, name, gender, age, input image and the predicted value after prediction and date of detection of input image.

1.6 Overview of the System

For this system (MDS) two main components are developed in this project frontend and backend. In backend pre-trained DL model DenseNet-121 is loaded into a web for detection which processes the skin images and it categorizes images in melanoma and non-melanoma after detection. Another component is frontend which allow users to enter their details such as Name, Age, and Gender and upload the input image for detection. The system is link with MySQL database for storing patient records and their results. Furthermore, the system have a good user-friendly interface through this users can easily interact with and perform detection by entering their details and input image, after detection the system provides result of predicted image.

Table 1.1: Chapters and their description

Chapter	Description
Chapter 2: Literature Review	In this chapter we discussed the importance of early detection of MSC and also different pre-trained models with custom CNN model are discussed.
Chapter 3: System Development Methodology	In this chapter we discussed the development process of this system.
Chapter 4: System Architecture and Implementation	In this chapter we discussed the system architecture like the frontend and backend designs, requirements for system implementation and also we discussed that how the system and model implementation is done.
Chapter 5: Results and Discussion	In this chapter, the results of pre-trained models and one custom CNN model with the model accuracy and model loss are discussed with graphs.
Chapter 6: Conclusion and Future Work	In this chapter conclusion, summary and achievements of the project are discussed as well as recommendations for future work.

CHAPTER 2

LITREATURE REVIEW

2.1 Introduction

Melanoma, which originates in the skin's melanocytes—cellular components that produce pigment is among the worst types of skin cancer. Skin cancer is the leading cause of death associated to skin cancer, even though its incidence is lower. This is due to its rapid spread to other parts of the body if detected too late. Melanoma arises mostly from damage to the DNA of skin cells caused by ultraviolet (UV) radiation from tanning beds or the sun. Early detection is critical for melanoma because the disease is highly curable when discovered in its early stages. That being said, once it spreads, the chances of survival drastically diminish.

Melanoma can appear anywhere on the skin, the face, arms, and back are most susceptible due to prolonged sun exposure. Some melanoma varieties might manifest in less obvious areas, like the soles of the feet or under the nails. Fair skin, a lot of sun exposure, a family history of the disease, and having a lot of moles are risk factors for melanoma. Good early detection methods are essential to improving the prognosis and survival rate for people with melanoma.

2.2 Importance of Early Detection in Melanoma

Early detection of melanoma raises the chance of a full recovery. Research indicates that in many cases, surgery is the only effective treatment for melanomas in the early stages that have not spread to other parts of the body. Above 90% of melanoma cases can be treated successfully if it is early detected. However, it will be so difficult if it reaches to other parts of body. This highlights that it is so important to early discover to take quick medical action for treatment.

Dermatologists use manual methods to physically and visually perform analysis of melanoma patients for melanoma, manually early detection of MSC is so challenging, even for professional medical experts. By using AI based and DL solutions developed for early detection of MSC we can get results so quickly and accurately. Through these technologies we can speedily detect the MSC early in limited time and even these are lifesaving and we can take medical actions quickly.

2.3 Deep Learning in Medical Image Processing

2.3.1 Overview of Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are well-suited for medical areas such as melanoma detection since they have reformed image processing activities. CNNs use and consists on several layers of filters to extract features like edges, textures, and forms from images, and they automatically analyze and learn from images to recognize patterns in those images. Because of this, CNNs are very best at analyzing arduous medical images, such as dermoscopic images of skin lesions.

In detection of MSC, CNN's are trained with the large dataset and datasets are divide into two sub directories in which one directory is kept for testing and another for training and both directories further divided into two sub directories one is for melanoma and another is of non-melanoma and then during training the model easily differentiate between melanoma and non-melanoma after training.

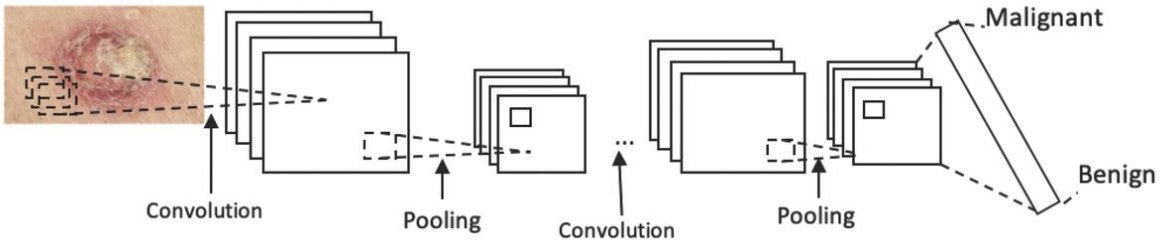


Figure 2.1

2.3.2 DenseNet-121

DenseNet-121(Densely Connected Convolutional Network 121) is a CNN architecture, this architecture performs remarkably well in medical image classification tasks. DesnesNet-121 architecture is consist on total of 121 layers in which all layers such as convolutional layers, pooling layers and fully connected layers are included. And this architecture is consist on dense blocks, every dense block is consist on convolutional layers in dense blocks every layer is directly connected to every other layers forwardly and this process promotes feature reuse and also reduce the risk of vanishing gradients. Between each dense block, transition layers are utilizes for downsampling and it uses the combination of Batch Normalization, 1x1 convolutions Average pooling layers with non-linear activation function ReLU. These layers (Transitional Layers) helps in reducing the dimensionality and it controls feature maps to

prevent increase in number of parameters. After last dense block this networks performs Global Average Pooling for reducing the spatial dimensions of feature maps to 1x1 feature vector. And fully connected layers with a sigmoid activation function is used to provide the output. This model have strong performance and efficient design and widely used for various tasks like medical image classification and object detection etc.

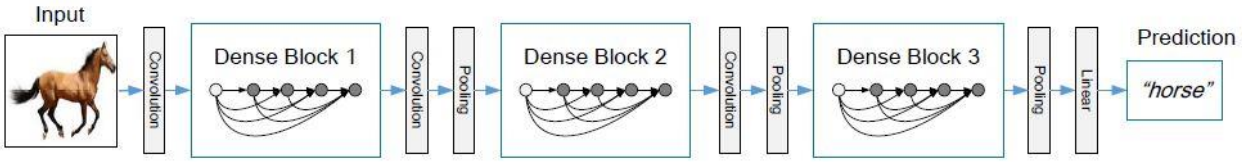


Figure 2.2

2.3.3 MobileNetV2

MobileNetV2, is a lightweight CNN architecture, designed specifically for embedded vision applications and mobile. MobileNetV2 architecture consists of convolutional layers followed by key features of MobileNetV2 architecture such as depthwise separable convolutions, inverted residuals, bottleneck blocks, linear bottleneck blocks and squeeze-and-excitation (SE) blocks and it uses ReLU activation function. These elements works together and reduce the numbers of parameters and computations required in maintaining the model ability in capturing complex features. This architecture is used because MoibleNetV2 is computationally efficient and lightweight and its performance is so good and provide high accuracy on medical image classification.

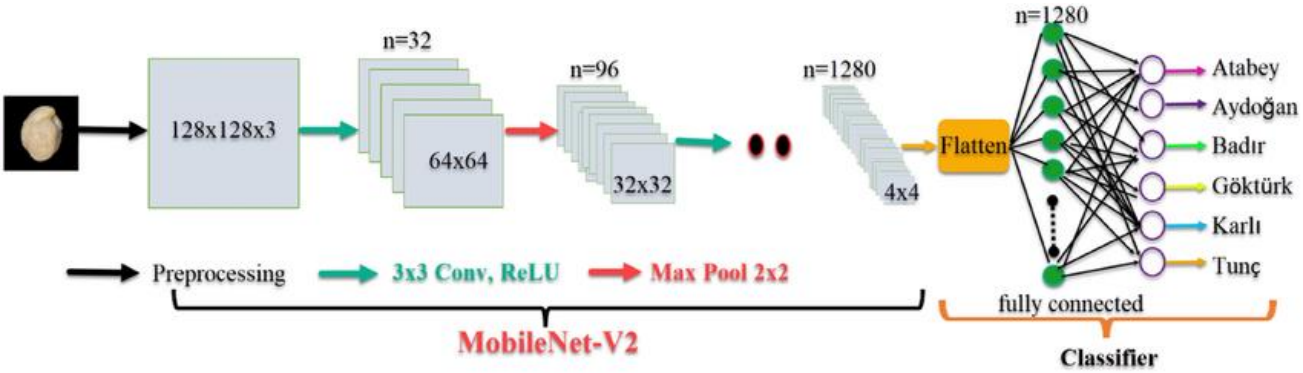


Figure 2.3

2.3.4 VGG-16

VGG-16 is a popular DL model for image classification proposed by Visual Geometry Group (VGG) at the University of Oxford. This model is consist of 16 convolutional layers and 3 fully connected layers, analyze and recognize features from images. VGG-16 uses 3x3 size filters in each convolution layer followed by max-pooling layer for extracting the features and finding and analysis of patterns in images.

With fully connected layers ReLU activation functions is used, in last dense layer sigmoid activation function is used for image classification between two classes and softmax for multiple class's image classification. In MSC image classification we use sigmoid function to differentiate between melanoma and non-melanoma. It is more powerful but slower than MobileNetV2 model, and it needs a lot of processing power and memory.

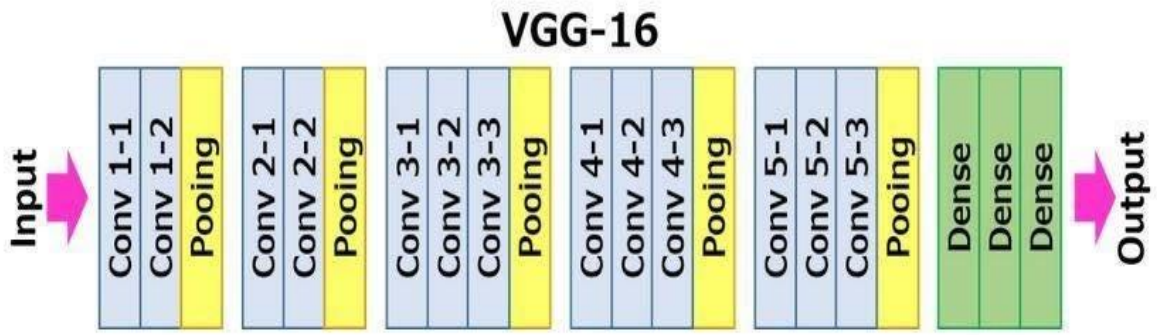


Figure 2.4

CHAPTER 3

SYSTEM DEVELOPMENT METHODOLOGY

3.1 Introduction

The system development methodology serves as the backbone of this project, guiding us step by step toward building a robust MDS. The methodology adopted focuses on leveraging DL models, including **DenseNet-121**, **MobileNetV2**, and **VGG-16**, for the early detection of MSC. Additionally, we integrated DenseNet-121 model into an easy-to-use web interface tailored for healthcare professionals and patients, allowing them to upload images and receive instant predictions.

The structure methodology ensures that each phase contributes effectively in the project goals. This chapter outlines the methodology we used, explain SDLC model and details the different phases we followed from planning to deployment.

3.2 SDLC Model

We adopted the agile methodology for this project, which is a good approach for software development by breaking down the whole project into small chunks. Agile model was selected because instead of waiting months to deliver the final product, agile model deliver the work after every phase if any changes need to adapt before moving to next phase. Through this model the project stay on track and can easily adapt to new changes in every phase if needed.

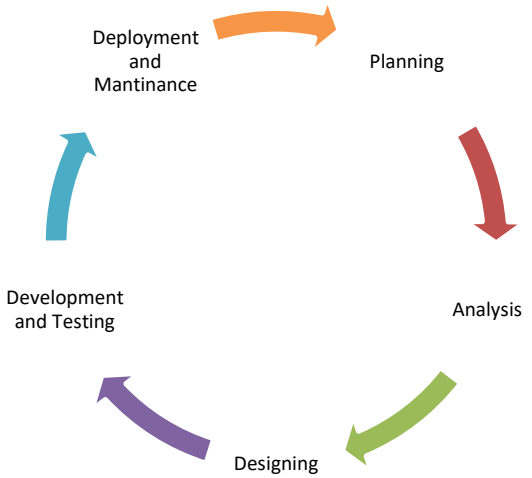


Figure 3.1

The Planning, Analysis, Design, Development, Testing, and Deployment phases are among the crucial phases in the agile paradigm. Each of these phases was essential in making sure our system maintained high accuracy and dependability while satisfying the functional and non-functional objectives. We describe the application of each phase to our MDS below.

3.2.1 Planning

The planning phase laid the groundwork for the entire undertaking. We described the objectives, constraints, and components required to build a functional MDS. Our top goals at this time were figuring out the datasets and tools needed to activate the system, as well as setting clear project objectives and deadlines.

Key tasks in this phase included:

- **Defining the Project Scope:** The project scope comprises creating a system that employs DL models to identify melanoma and creating an intuitive web interface where patients and clinicians may upload photographs and receive predictions.
- **Defining Objectives:** Our goal was to create a system that could identify melanoma early on by merging the DenseNet-121, MobileNetV2, and VGG-16 models.
- **Allocating Resources:** Image collections (such as the kaggle dataset) and technical instruments such as TensorFlow for DL models, MySQL for database administration, and Django for the backend were discovered.
- **Timetable and Reference Points:** Important turning points were preprocessing the dataset, training the models, integrating the models with the online interface, and carrying out thorough testing prior to the final deployment.

3.2.2 Analysis

Throughout the analysis phase, we carefully looked at the system's technological requirements. The functional and non-functional requirements of the MDS were identified at this phase, and every aspect, including data preprocessing and user interface design, was thoroughly thought out.

Key tasks in this phase included:

- **Functional Analysis:** We focused on the core functionalities of the system, such as image submission, DL model detection of melanoma, and result presentation. We also considered how the system would handle multiple users and their personal data.
- **Data analysis:** To ascertain whether preprocessing stages, like augmentation,

normalization, and scaling, were required, the image datasets were analyzed.

- **Model Selection:** Among the DL models we researched and looked at were DenseNet-121, MobileNetV2, and VGG-16. We choose which models to utilize by setting goal of two things, one is performance and another is accuracy.
- **Database Requirements:** To store the user data, such as inserted input image, and patient details such as patient-id, name, age, gender and the predicted image result after prediction. This functionality is so important because of saving the patient record. And MySQL databases was supposed to store the patient data.
- **User Interface Requirements:** We analyze the need of a simple and easy online interface for both Medical professionals and patients to utilize easily for skin cancer images detection. The user interface has to be simple to use, facilitate a quick submission of images, and clearly show the outcomes.

3.2.3 Design

During the design phase, an architecture blueprint containing the frontend and backend components of the system was established. The goal was to create a unified system in which users could simply engage with the DL models via an online interface, with backend processes managing data storage and predictions.

Key tasks in this phase included:

- **User Interface (UI):** We developed an online interface that allows users to add images, view prediction results, and manage their data. Django framework is used for developing responsive and easy to use user interface for inserting images and receiving results of predictions.
- **System Architecture:** The Architecture of System made by using full-stack web application. Django is used for building the backend by integrating the DL models (DenseNet-121, MobileNetV2, and VGG-16) for predicting MSC images for melanoma detection. MySQL database is used for safely storing the user data and input images and results after prediction of input images. Further, to control communication between the frontend and backend, REST API endpoints were developed for this process.
- **Models for DL:** We designed and trained the VGG-16, MobileNetV2, and DenseNet-121 model integration for the system. These algorithms were trained using a melanoma skin images dataset in order to provide accurate prediction results and improved its performance. Each model was configured to analyze the images accurately and generate a confidence value after melanoma predictions.

3.2.4 Development and Testing

In this phase the actual system was built, in this phase the development of web interface and the connectivity or linking of database with interface was performed, and also the integration and implementation of pre-trained model DenseNet-121 after some improvement in performance with a system were performed. In this phase the goal was to create a functional product.

Key tasks in this phase included:

- **Model Development:** Model development was the core process in which different 3 pre-trained models such as DenseNet-121, MobileNetV2, and VGG-16 were trained and a custom CNN model was developed and trained with the same dataset of melanoma and non-MSC images.
- **Frontend Development:** In frontend development, Django framework was used to develop a responsive online interface, through which users can enter their data such as name, age, gender and upload the image for prediction of MSC and also will show the results with the confidence value after prediction. And this interface will be find so easy and simple to interact with by both health experts as well as patients.
- **Backend Development:** In backend development, a pre-trained model DenseNet-121 was implemented and integrated with system for prediction of skin images to predict melanoma, this model has a good accuracy and perform nicely by providing accurate result in detection. Also MySQL databased was created for storing patient information and uploaded images and also the results after prediction of images with the confidence value. For safety of the system and patient data we included the user authentication to protect the system and patient data.
- **Testing:** An essential step in the development process was testing. Unit tests were carried out on separate parts of the system, like image uploading and prediction processing, while integration tests made sure the system worked properly overall. In order to make sure the models fulfilled the project's objectives for early melanoma detection, we also assessed the models' performance and accuracy.

3.3 Dataset Preparation

3.3.1 Data Collection (18807 Images from Kaggle)

A dataset of 18807 high-quality images of skin lesions was obtained from Kaggle and employed in the development of the MSC detection system. For hosting datasets used in machine learning and DL research, particularly in domains like medical imaging, Kaggle is a well-known website. In order to train the model to distinguish between melanoma and non-melanoma skin disorders, the dataset contained images of both melanoma (malignant) and non-melanoma (benign) skin lesions. Using a Kaggle dataset expedited the data collecting procedure because there were so many publicly available, pre-labeled images. The dataset included a fair representation of different skin types, lesion sizes, and textures, all necessary for developing a trustworthy MDS. By using the Kaggle dataset, we ensured that our model was exposed to a variety of real-world circumstances, which enhanced the system's generalization abilities.

3.3.2 Preprocessing of Images (Resizing, Normalization)

The images needed to be preprocessed in order to be ready for model training after the dataset was gathered. Because the images in the dataset varied in terms of size and quality, preprocessing made sure the data was consistent and prepared for the DL models to analyze it effectively.

- **Resizing:** We converted all of the images from the Kaggle dataset to a fixed size of 224x224 pixels because they were provided in a variety of sizes and dimensions. This resizing was required to comply with the standard image size criteria of the DL models, namely DenseNet-121 and VGG-16, for input. Because it maintains computational efficiency during training while preserving crucial features in the skin lesions, the 224x224 resolution was chosen.
- **Normalization:** The images pixel values were adjusted following resizing. Pixel values in the original dataset were scaled down to a range between 0 and 1 for best model performance. Originally, pixel values spanned from 0 to 255. By mitigating the effects of fluctuating illumination and image contrast throughout the dataset, normalizing the pixel values aids in the model's more effective learning.
- **Data Augmentation:** The process of applying different alterations to the original pictures of a training dataset is known as data augmentation. These transformations consist of actions such as moving, flipping, rotating, zooming, and modifying contrast or brightness. The goal of data augmentation is to produce different versions of the

current pictures so that the model may learn from a greater range of patterns and improve its ability to generalize. This minimizes overfitting and enhances the performance of the model, particularly in cases when the dataset is small or unbalanced.

3.4 Deep Learning Model Selection

The right deep learning model needed to be selected in order to correctly diagnose melanoma. We assessed several models, focusing on their ability to identify images of medical problems. Following are the models which are tested and reason behind our choice.

3.4.1 DensesNet-121 for Melanoma Detection

For melanoma detection we select and choose DenseNet-121 as our main model because of its accuracy and efficiency in image classification tasks, specially when it comes to medical images. Because of its distinct design, which connects all of the layers, DenseNet-121 improve learning and reduce the risk of overfitting because of reusing features. Furthermore, it nicely uses fewer parameters more effectively, it maintain the accuracy without using so much computation power. We improved the DenseNet-121 model using the kaggle melanoma skin images dataset to achieve the high accuracy.

3.4.2 Other Experimented Models (MobileNetV2, VGG-16)

- **MobileNetV2:** MobileNetV2 is a good pre-trained low weight model and best to use in web-based interfaces and mobile devices. Although it is fast and effective, but it didn't perform well as correctly as DenseNet-121, but this model is still a best choice to use because of its minimum resources usage.
- **VGG-16:** VGG-16 is an eminent model because of extracting complex features from images. But it need much computational power because of its more parameters, its performance was no so good. By comparing VGG-16 with DenseNet-121, DenseNet-121 perform amazing and achieve high accuracy.

  Figure 3.2 shows the process from data collection to prediction.

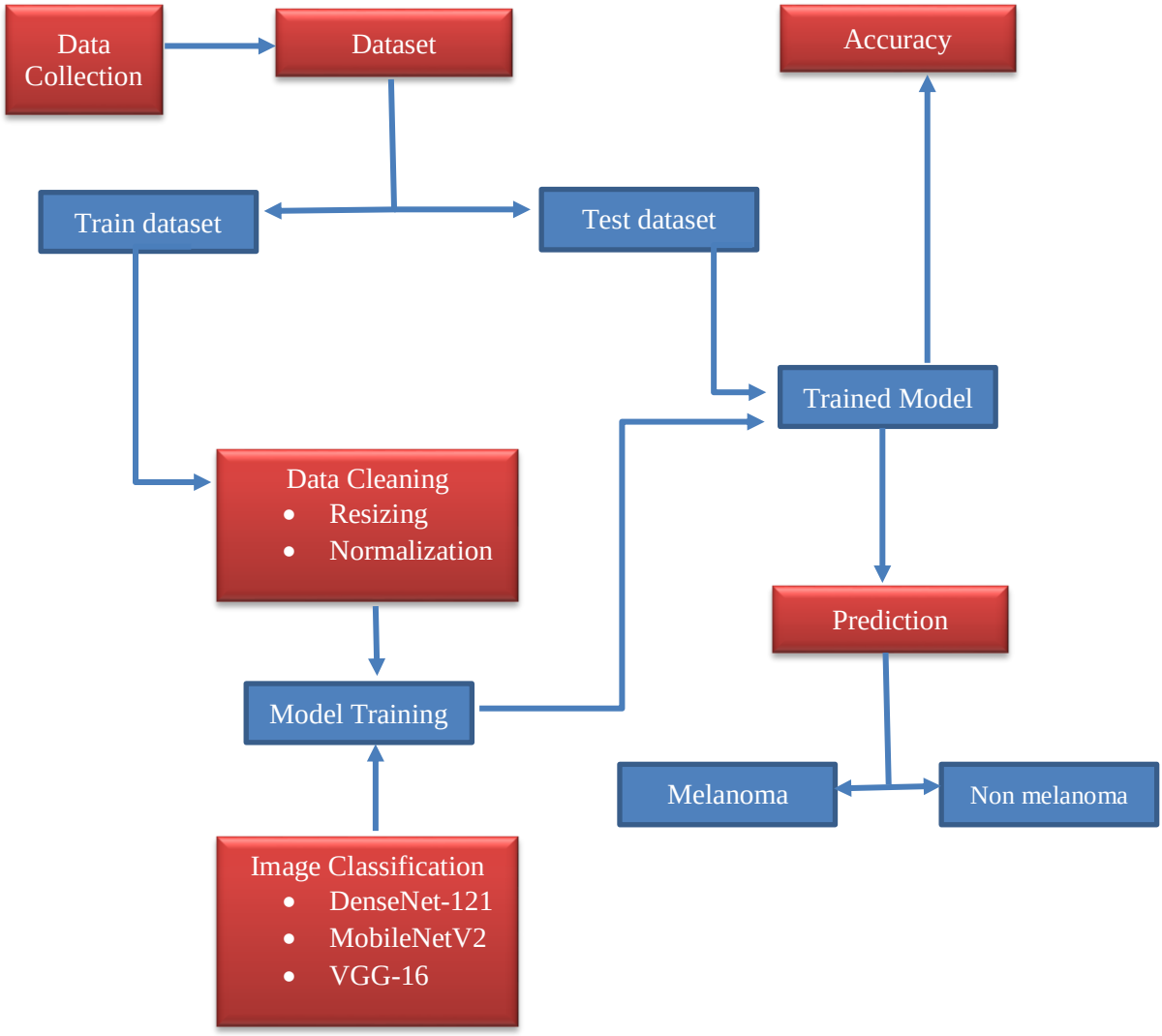


Figure 3.2

3.5 User Interface (UI) Design and Development

User interface is the main part of melanoma detestation system, and it makes sure that patient and healthcare experts can easily interact and use the system. The user interface was create to easily upload images of skin, to see prediction results and to interact with the system. To give or provide easy user interaction with system, the interface should be visually appealing, user-friendly and responsive. And also the interface has to be simple navigation bar to navigate easily. The main goal of developing and designing user interface was to make a system more visually appealing, easy to use, interactive and easy to understand.

3.5.1 Frontend Development Using Django

The frontend of MDS was developed or created by using Django, which is a python framework for developing websites. This framework is robust and facilitates quick development. All web pages of the websites and its templates was created with the help of Django. The user interface of the system is responsive for every small and large screens such as laptops, mobiles, tablets and other screens.

Following are the important features created in the frontend are include:

- **Image Uploading:** A prediction page has an easy to use form for entering patient details and for uploading images.
- **Prediction representation:** When we uploading image for prediction, the image is preprocessed and then the result of prediction with the confidence value and expected result that can be melanoma or non-melanoma are displayed in understandable and clear manner.
- **Navigation:** Easy navigation that facilitates users' movement between the systems many sections, such as the prediction page, about page and contact page.

3.5.2 User Interaction Flow

In order to make the process as easy and effective as possible, the MDS's interaction flow was developed. Because of the system's straightforward and sequential steps, users may be assured of a seamless experience from uploading an image to receiving the melanoma forecast. The following is a description of the main user interaction flow:

- **Upload Image:** After visiting the melanoma detection website, users are asked to upload a picture of a skin lesion. To prevent confusion, the interface clearly indicates what sizes and types of images are permissible.
- **Submission for Prediction:** When users click the "Submit for Prediction" button after uploading an image, the backend processes the image using a DL model that has already been trained (DenseNet–121).
- **Real-Time Feedback:** A progress indicator or status message alerts users that the forecast is being generated while the system processes the image.
- **Prediction Outcomes Display:** The anticipated diagnosis appears on the screen following the model's analysis of the image. The outcome comprises the likelihood of the lesion being benign or malignant, as well as the probability of prediction.

3.6 Backend Integration

The processing of user inputs, saving of important data, and facilitating seamless communication between the frontend (user interface) and the underlying DL models are the responsibilities of the backend in our MDS. All data, including user photos and predictions, are efficiently managed, saved, and retrieved thanks to a strong backend system. This section explains the essential components of the backend integration, focusing on the usage of MySQL for data storage and the image upload and result retrieval operations.

3.6.1 MySQL Database for Storing Results

This project uses MySQL as its database management system to store user data, uploaded photos, and related prediction results. Because of MySQL's dependability, scalability, and simplicity of integration with Django the framework that powers our application's front end and back end it was chosen.

We made several tables in the database to hold the following crucial data:

- **User Data:** This comprises the standard user data, like email addresses, encrypted passwords, and usernames. The MySQL database is connected with Django's built-in authentication mechanism to manage user credentials and access control.
- **Image Data:** This table maintains information about the images that users provide for melanoma detection. It keeps track of the uploaded photographs' file paths, the user who uploaded them, and metadata like the upload date and time.
- **Prediction Results:** This table contains the prediction confidence and the outcome (either "melanoma" or "non-melanoma") of the DL model's processing of an image. Users can see their past forecasts and monitor changes over time thanks to this.

3.6.2 Image Upload and Result Retrieval Process

Through the web interface, users can upload images of skin lesions to the system in order to diagnose melanoma. After an image is submitted, it is processed by the backend using DL models that have already been trained. The image and prediction result are then stored in a MySQL database, and the user receives the outcome. Following are the steps how procedure operates:

- **Upload Image:** In this step users upload images of skin lesions through user friendly interface, and file handling mechanism of Django is used for sending images to backend for processing.
- **Preparation and Initial Processing:** In this step the upload image is resized, normalized and augmented and adjusted to the model requirements for prediction. After

uploading and preprocessing of image the preprocessed image, is then given to a trained model which we chooses , that is DenseNet-121 to detect melanoma. Then after prediction, the predicted confidence value and the prediction outcome, either may be ‘melanoma’ or ‘non-melanoma’ are generated by the model.

- **Keeping the Results Stored:** MySQL database is used for storing the uploaded image and the prediction results for later use.

CHAPTER 4

SYSTEM ARCHITECTURE AND IMPLEMENTATION

4.1 Introduction

In this chapter, we will see details about implementation and architecture of the MDS. The system was designed with an easy to use interface and DL model was integrated with a strong backend. We will see the system's hardware and software needs and requirements as well the flow of architecture to ensure effortless functioning. This section gives a general overview of the system architecture.

4.2 Overview of System Architecture

The system is consist of many components, each component of which adds to the overall usability and functionality in the system. Essentially, system uses the trained model to predict melanoma. It is housed on a web platform powered by Django and uses a MySQL database to handle user information and outcomes. The system's structural flow and technical characteristics are examined below.

4.2.1 Hardware and Software Requirements

The MDS requires specific hardware and software settings in order to operate correctly. These criteria cover the development environment as well as the deployment settings.

Hardware Requirements:

A development system equipped with a powerful CPU or GPU (NVIDIA GPU, for example, for quicker model training).

For managing big datasets and DL model training, at least 16GB of RAM is required. Enough storage space (100GB minimum) to store user-uploaded photos, model checkpoints, and datasets.

Internet access to cloud-based services (pre-trained models, Kaggle datasets, etc.).

Software Requirements:

- **Programming language:** Python is the preferred programming language for backend development, with TensorFlow/Keras available for model implementation.
- **Frameworks:** Django, which helps with model integration and user interaction for the web application.
- **Database:** MySQL to hold user information, such as uploaded photos and forecast outcomes.
- **Libraries:** TensorFlow, Keras, NumPy, Pandas, and Matplotlib are libraries used in

data processing and DL. Web development libraries built with Django.

- **Cloud services (optional):** Google Colab, which enables the utilization of cloud-based GPU resources for model training and testing.

4.2.2 System Flow and Structure

The system functions in a systematic way from the moment the user uploads a photo until they receive the melanoma prognosis. The system flow and structure shown as below in diagram:

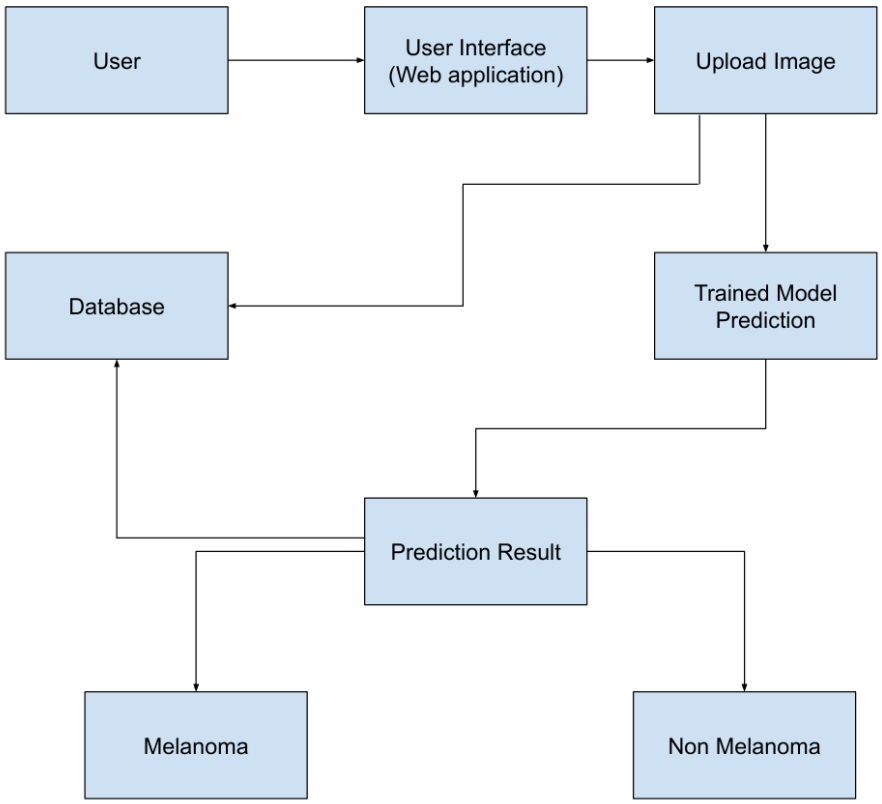


Figure 4.1

The following are the components that make up the overall structure:

i. User Interaction (Frontend):

Through the web interface, the user can upload pictures of skin lesions to be used in the detection of melanoma. Developed with HTML, CSS, and Django templates, the frontend offers a smooth and user-friendly experience.

ii. Image Upload:

When an image is uploaded, it is sent to the backend for preparation. This means that the image must be reduced in size and normalized to match the required input size for the DL model.

iii. Model Prediction:

The DL model (DenseNet-121) receives the preprocessed image and generates a forecast. There is a likelihood score and two possible outcomes: "melanoma" or "non-melanoma," respectively. For the sake of experimentation and comparison, additional models like MobileNetV2 and VGG-16 are also integrated.

iv. Result Storage:

The user-uploaded image and the prediction's outcomes are kept in the MySQL database. Users can follow their history of melanoma risk and view their previous predictions thanks to this.

v. Result Display:

The frontend presents the user with the prediction and confidence score.

4.3 Frontend Design

The MDS's frontend design is essential because it makes sure that users both patients and medical professionals can communicate with the system effectively and conveniently. The user interface should to be interactive, easy to use or user-friendly, and responsive in using for users.

4.3.1 UI/UX Design Considerations

The main goal of the UI/UX design is to ensure that users utilizes the system with ease, without technical expertise. Therefore, UI/UX design gave attention to the following impressive elements:

i. Simplicity:

The UI was proposed to be appealing and simple, the buttons and sections was attractive and easy to function and understandable. This ensure that users won’t face difficulties by using the system components. .

ii. Responsiveness:

Due to responsiveness the system was designed to responsive in functioning by using in

multiple screens such as mobiles, laptops, desktops, tablets and other screens, to easily use by users with any screen.

iii. Accessibility:

The system was designed helpful with accessibility features like easy to use buttons, and easy to use interface with larger easy to see text, because users may have different levels of eyesight and no skills and experience of using digital systems.

iv. User Feedback:

Feedback is available instantly for every interaction with the system. For instance, users see a confirmation message when they upload a picture. In a similar vein, the outcomes of a prediction are shown instantly, and the processed image viewing option is also provided.

v. Consistency:

The MDS's branding is consistent with the color palette, typography, and overall design. White and blue were selected to create a polished, clean aesthetic that reflected the gravity of melanoma discovery.

4.3.2 Navigation and Functionality of the User Interface

The system's functionality and navigation were created with the goal of requiring the least amount of effort from users to accomplish their jobs. When developing the interface, the following factors were taken into account:

i. Home Page:

The main page acts as the system's entry point. After receiving a brief introduction to the system as shown in figure 4.2. Users can choose to register (for new users) or log in (for current users) as shown in figure 4.3 and figure 4.5.

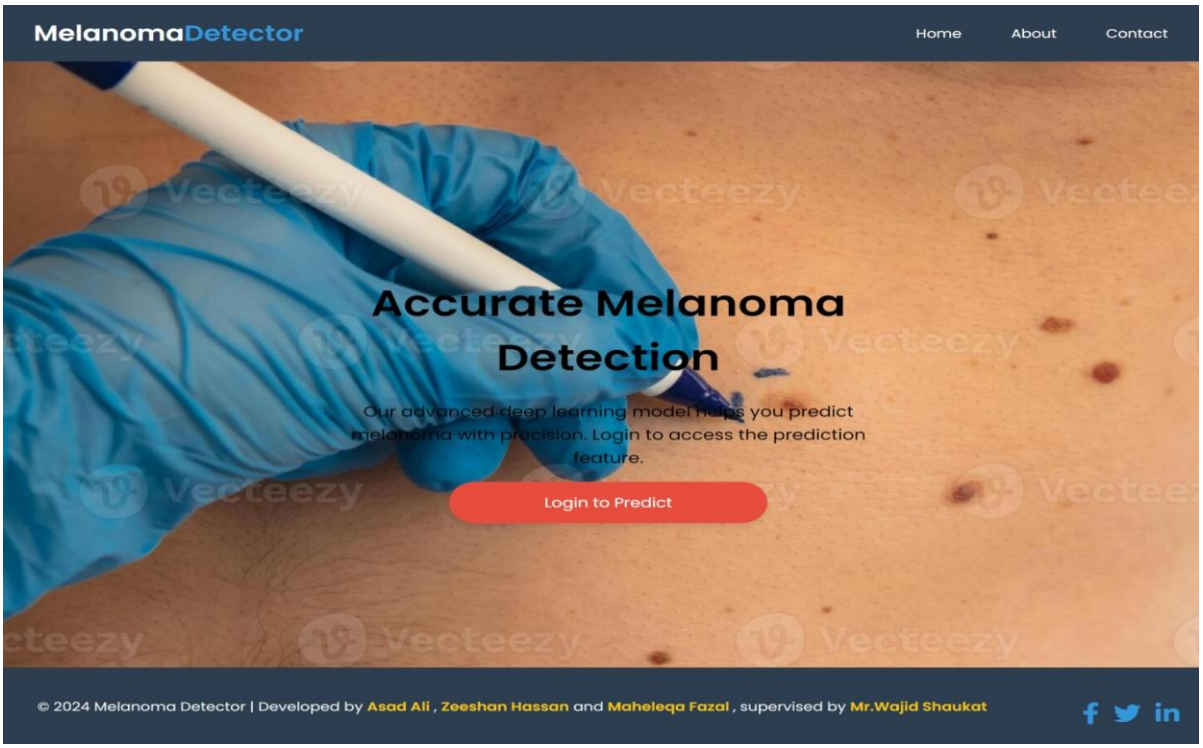


Figure 4.2

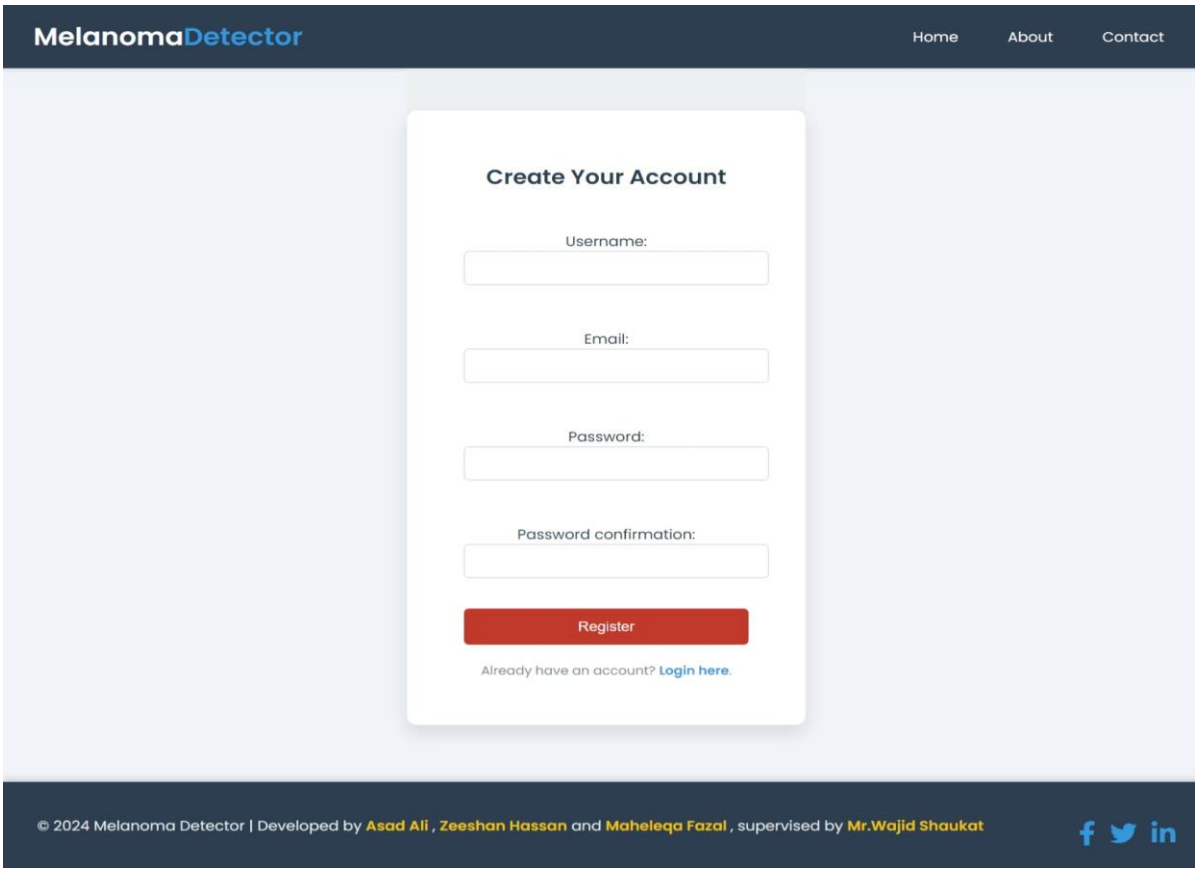


Figure 4.3

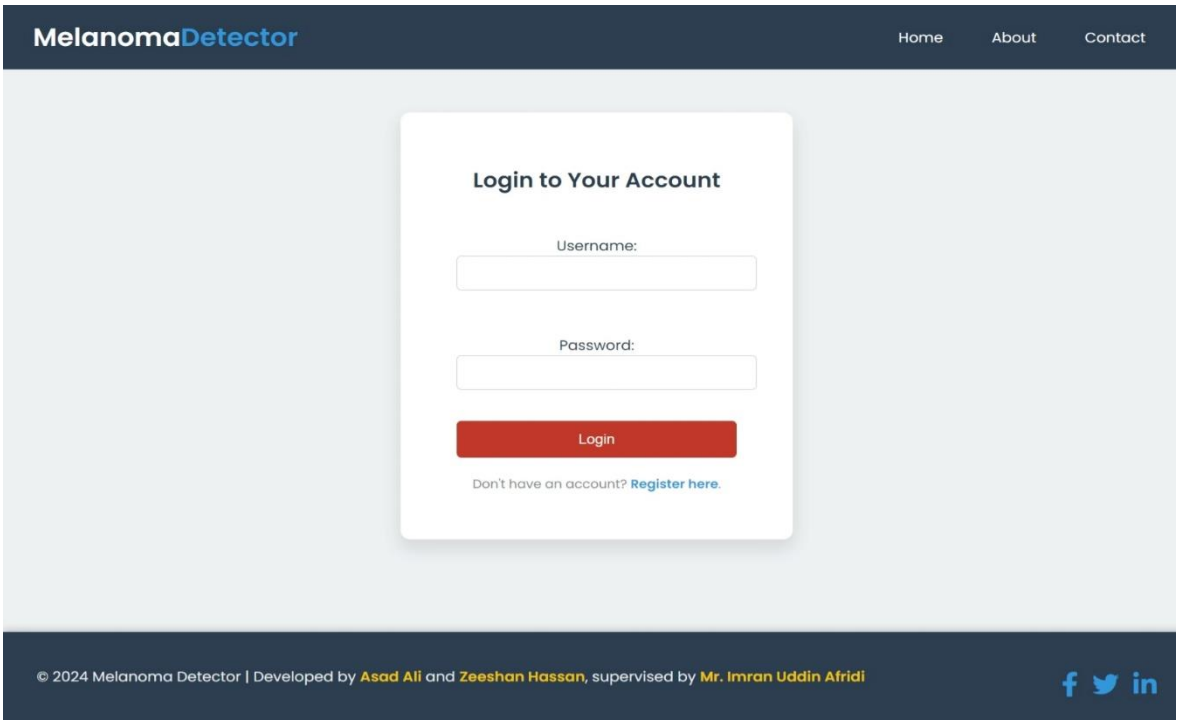


Figure 4.4

ii. Patient details and image uploading page:

The main use of this page is to allow users to enter patient details such as patient id, name, age and gender then user upload skin images for the purpose of predicting melanoma. Using this function is made simple by the prominent display of the "choose file" button. Users can view the various processing stages (input image, preprocessing, and output) that go into producing the final forecast after uploading an image.

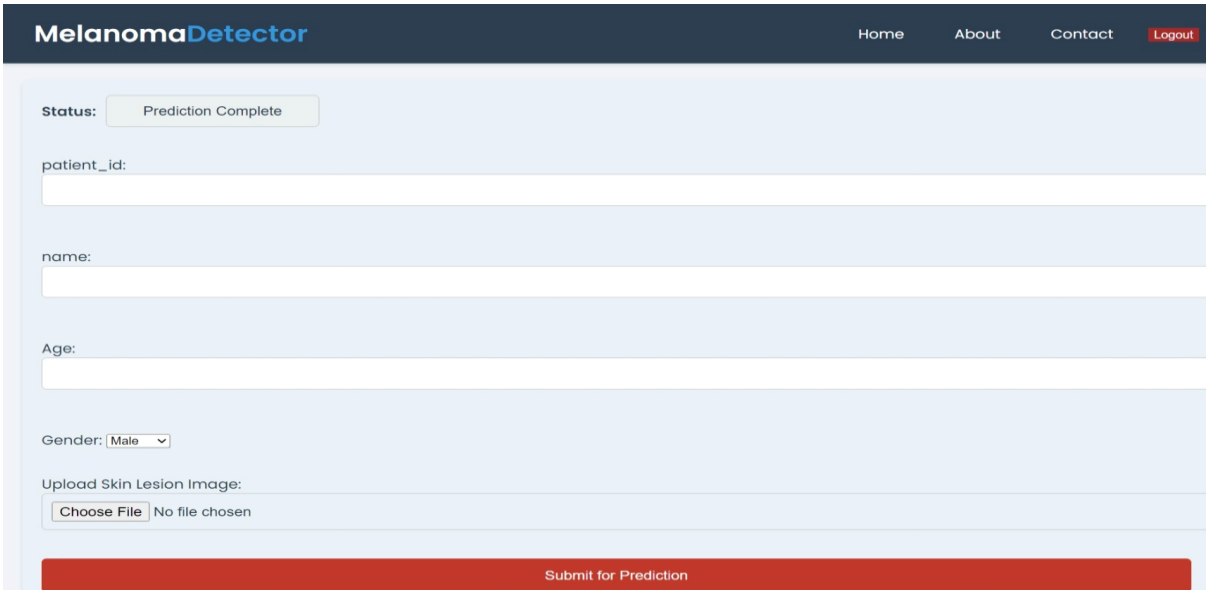


Figure 4.5

iii. Prediction Results Display:

After an image is processed, the results are displayed clearly. The interface shows whether the image is predicted to be melanoma or non-melanoma, along with additional information such as the confidence level of the prediction. All the details are shown in figure 4.6.

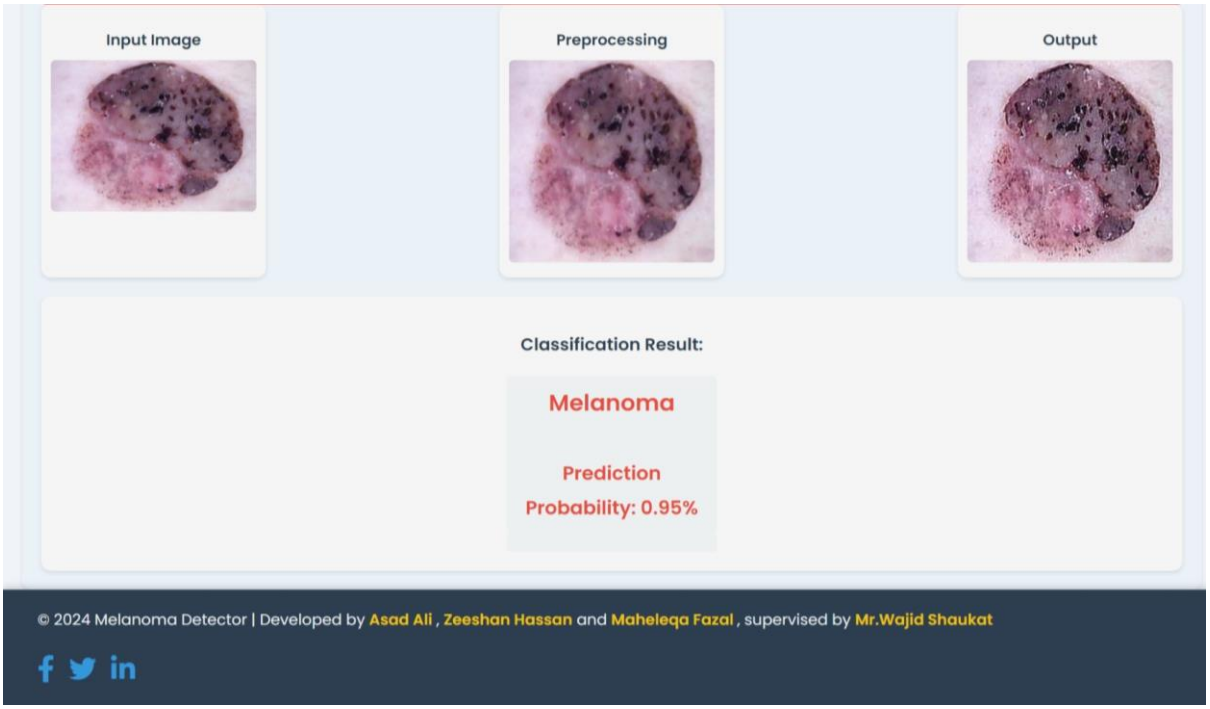


Figure 4.6

iv. Contact page:

Contact page will use to ask questions and make requests for contact. It has a basic form that users may fill out with a message and their contact information. This page makes sure that people can quickly access assistance when they need it. The interface is shown in figure 4.7.

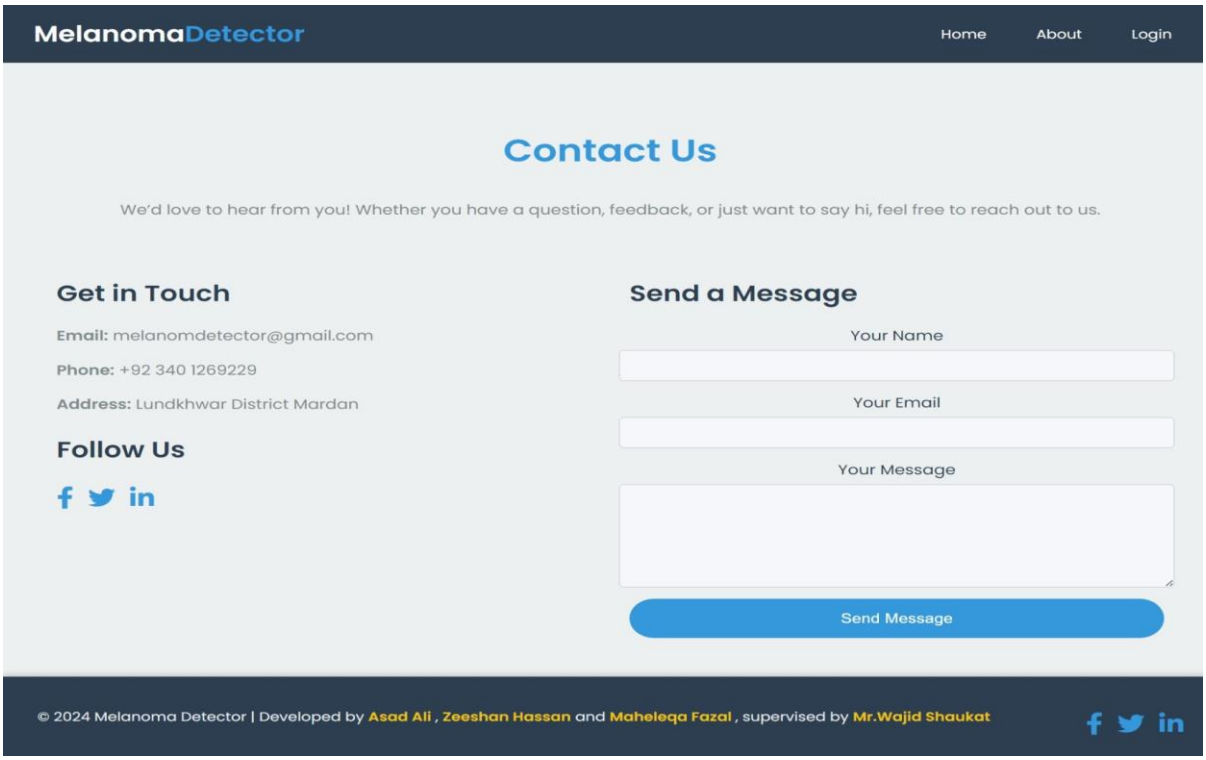


Figure 4.7

v. About Page:

The about page goes into great detail regarding the purpose of the system, the individuals who designed it, and the manner in which it aids in the early diagnosis of melanoma. Details are shown below in figure 4.8.

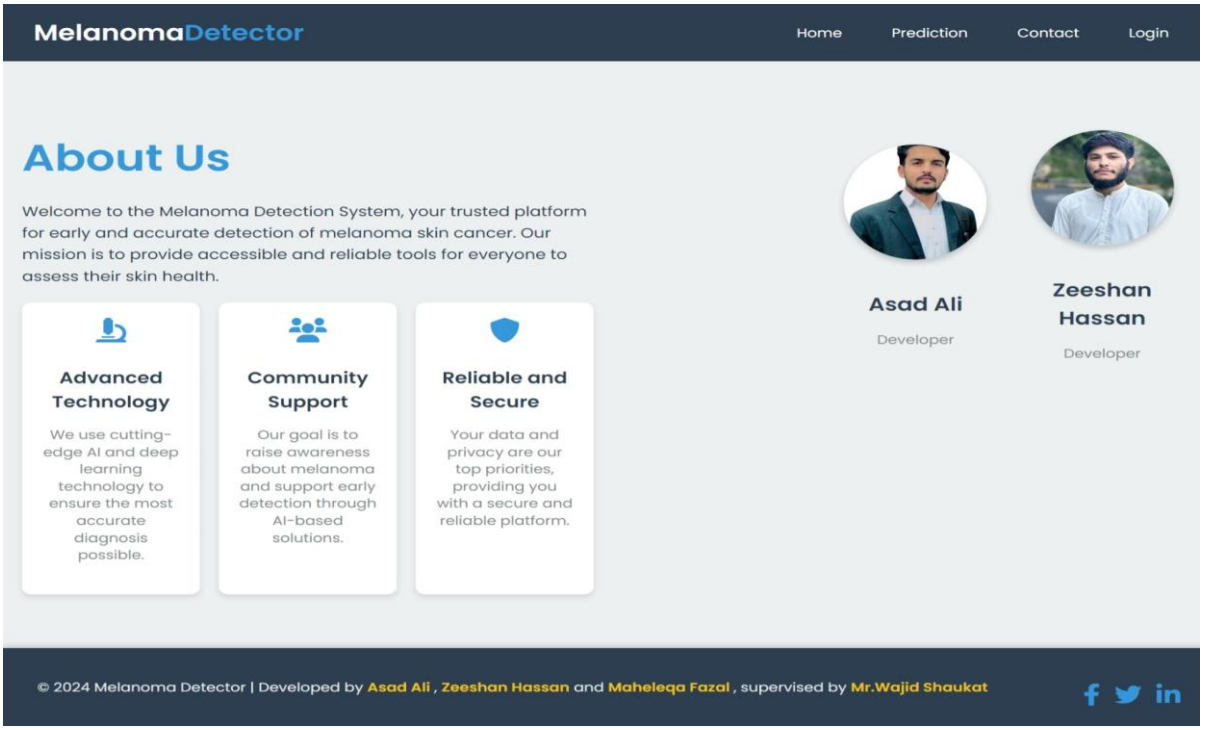


Figure 4.8

vi. Navigation Bar:

The navigation bar, which is fixed at the top of the screen, offers quick access to crucial functions including "Home," "About page" "Contact page" .Everyone can easily access the different pages of the website.



Figure 4.9

4.4 Backend Design

The primary component of the MDS is its backend, which manages image uploads, data processing, results archiving, and prediction communication back to the frontend. This system component ensures that the database, DL algorithms, and user interface all function together flawlessly to allow users to input photos and receive prompt, accurate melanoma forecasts.

4.4.1 MySQL Database Schema for Storing Results

The key components of the system, such as user data, picture metadata, and prediction outcomes, are managed and stored in a MySQL database. Because of its scalability, speed, quick response and dependability, MySQL database was selected.

Following components are included in database schema:

- **Table for user authorization:** This table holds user credentials, like passwords, email address, and username. This was so important for managing user authentication (login). This is so important to secure and protect users.
- **Patient Details Table:** This table is uses for saving uploaded images of skin lesions, patient details like patient-id, name, age, and gender. Also this table is used for saving the prediction results like prediction confidence level value the detected type which may be melanoma (malignant) or non-melanoma (benign) when model predict the skin lesion image and also keep track of the date when the image was predicted.

4.4.2 Image Upload and Data Retrieval

- **Image Upload:** Using the UI, users submit pictures of skin lesions. The image is uploaded by the system to a secure directory on the server after it is submitted. The file path is kept in the database together with other metadata (such as the upload date). By

doing this, the system is guaranteed to be able to quickly access and use the picture as needed.

- **Data Processing and Retrieval:** The backend uses DL models (such DenseNet-121 and VGG-16) to process the image after it is submitted. The outcome and the image data are both stored in the database when the model generates a prediction. The user is presented with an interface that displays the prediction result, which is obtained from the backend and indicates if the lesion is benign or malignant. Customers that have an account can also view previous forecasts.

4.5 Deep Learning Model Implementation

The core of the MDS consists of DL models that have been tuned and trained to obtain optimal precision in identifying skin lesions as either non-melanoma or melanoma. Several models, including DenseNet-121, were trained using a collection of images depicting the disease in order to identify the early stages of melanoma.

4.5.1 Training DenseNet-121 for Melanoma Detection

A state-of-the-art CNN known as DenseNet-121 was selected for this study due to its efficacious feature extraction capabilities and proven performance in medical photo categorization tasks. The DenseNet-121 model was pre-trained on the ImageNet dataset and then refined using a bespoke melanoma picture dataset from Kaggle. Key steps in the training process included:

- **Preprocessing:** To prepare the dataset for DenseNet-121, all photos were resized to 224 by 224 pixels, the minimum needed input size, and their pixel values were normalized.

- **Transfer Learning:** To identify skin lesions as melanoma or non-melanoma, custom layers were added to the last layers of the DenseNet-121 model after it was started with weights pre-trained on ImageNet.

- **Training:** The model was optimized using the Adam optimizer with a batch size of 32 using binary cross-entropy as the loss function. Techniques for augmenting data were also used to strengthen the model's resilience and avoid overfitting.

4.6 Testing and Evaluation

In this last phase of system implementation, the performance evaluation of DL model and the interface is involved in testing the integrated system.

4.6.1 Testing the User Interface and Model Integration

After training the models successfully, testing of interface between user interface and the backend was done. Through user interface which is built with Django, users can upload the skin lesion images and receive the prediction confidence level after performing prediction by DL models. This testing process ensured that:

- **Smooth Integration:** The DL models can process images and return predictions result accurately, and also the frontend can accurately upload images to backend smoothly.
- **User Experience:** The interface of system was tested to ensure that it is so simple and easy in using by both medical experts and patients and could upload images and examine the findings.

4.6.2 Performance Evaluation of the System

In terms of performance evaluation the MDS was evaluated in accuracy, speed and efficiency. Following were assessment metrics:

- **Model Accuracy:** In assessing the performance of DesnseNet-121 model, the accuracy and loss measures were used. The model was evaluated for accurate classification of unseen images of skin lesion, additional test dataset was used.
- **Prediction Time:** The system prediction time was monitored to ensure that the system could provide quick results to users in less amount of time.
- **System Usability:** To assess the usability of user interface we perform user testing and to get feedback from users that how easy the interface was in uploading images, how fine the design was in using, and how easy it was in viewing the prediction results.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter describes the MDS evaluation process, with a focus on the use of DL models for accurate and efficient MSC detection. In addition to a new DL model, this system was constructed using three pre-trained models (DenseNet-121, VGG-16, and MobileNetV2) and a custom CNN model. And all models are trained with same dataset, and dataset consist on 18807 images. In dataset 70% images are kept for training and 30% for testing. While training models, pre-trained models trained with 20 epochs and the costume CNN model is trained with 10 epochs.

Furthermore this section will discuss the training process, model performance in terms of detecting skills, and a comparative study to determine which model is optimal for our MDS.

5.2 Performance Evaluation and Results of Trained Models

A total of four models are trained in which 3 are pre-trained models and one is custom CNN model and each model is evaluated for accuracy and loss individually. Following are the trained models:

1. Custom CNN
2. DenseNet-121
3. VGG-16
4. MobileNetV2

5.2.1 Custom CNN

The custom CNN model was built with total of 4 convolutional layers and ReLU activation function is used followed by fully connected layers with sigmoid activation function in last layer to classify skin lesion images as either melanoma or non-melanoma. The custom model was trained using the dataset of total 18,807 images which was downloaded from Kaggle in which 70% images were kept for training and 30% for testing and model was trained with total of 10 epochs, with a focus on optimizing for melanoma detection. The architecture of CNN model is shown in Figure 5.1.

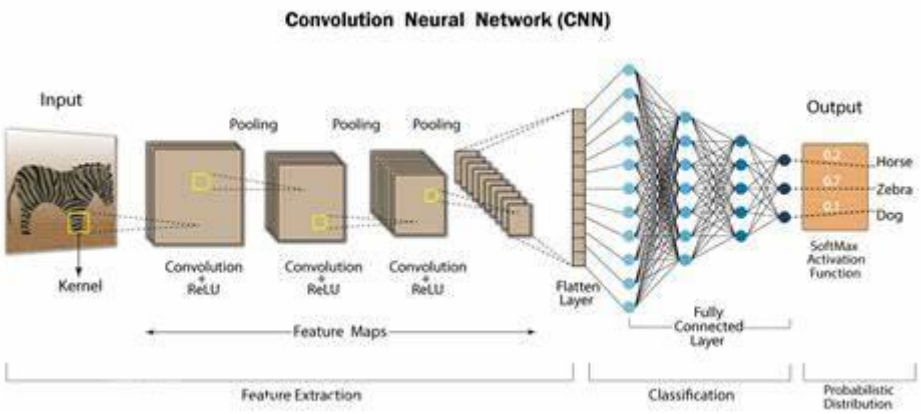


Figure 5.1

The training and validation accuracy is shown in figure 5.2

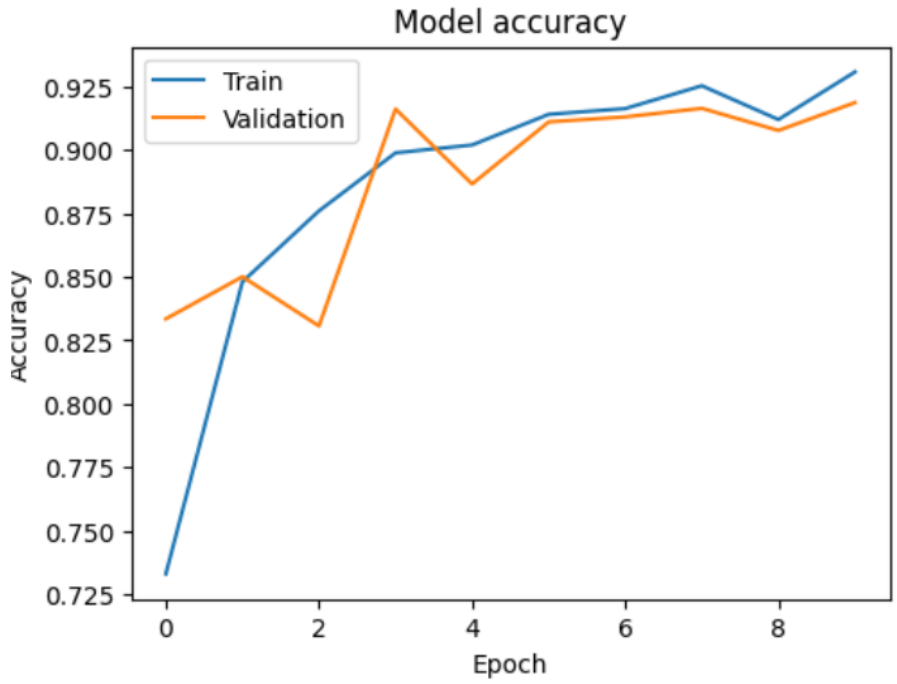


Figure 5.2

Figure 5.2 shows the training and validation accuracy of the custom CNN model. The training accuracy in starting was 64 and the validation value was began from 83. In each epoch the training and validation accuracy was improving, at the last epoch the model achieved 91% of accuracy.

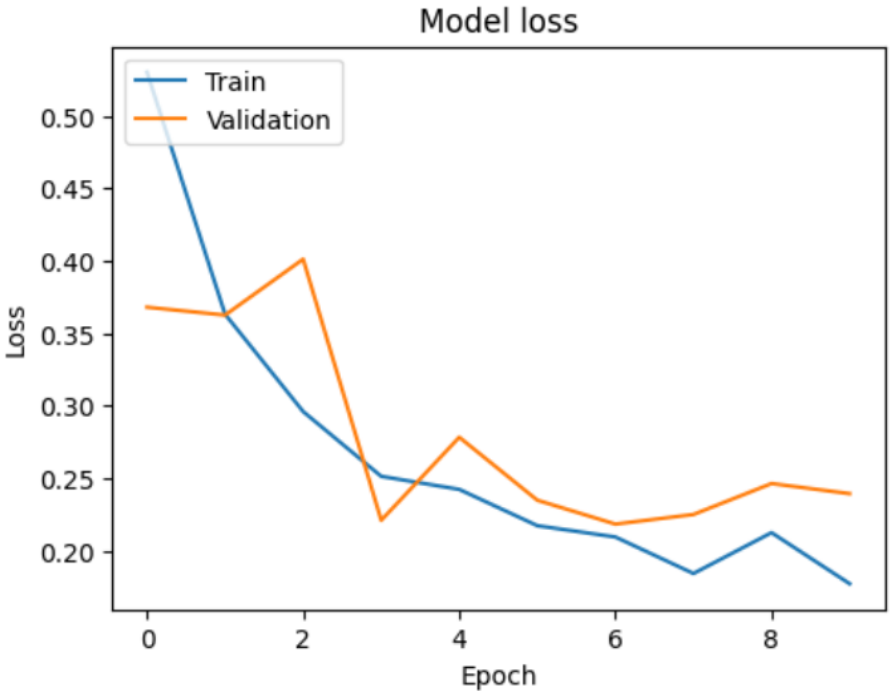


Figure 5.3

Figure 5.3 shows the model training and validation loss, the model training loss in starting was 62 and validation loss value was 36. During training the model training and validation loss was continuously decreasing, in last epoch the training loss value was 18 and validation loss value was 23.

5.2.2 DenseNet-121

The popular pre-trained DenseNet-121 model was modified to work with the melanoma dataset. This paradigm creates feed-forward connections between each layer and all other layers, allowing for efficient feature reuse and improved performance. DenseNet-121's ability to recognize minute details in images of skin lesions was enhanced by modifying its pre-trained weights for melanoma classification. ImageNet was used to train DenseNet-121 initially. The ability to identify melanoma was significantly enhanced by this approach. The architecture of DenseNet-121 Model is shown in figure 5.4

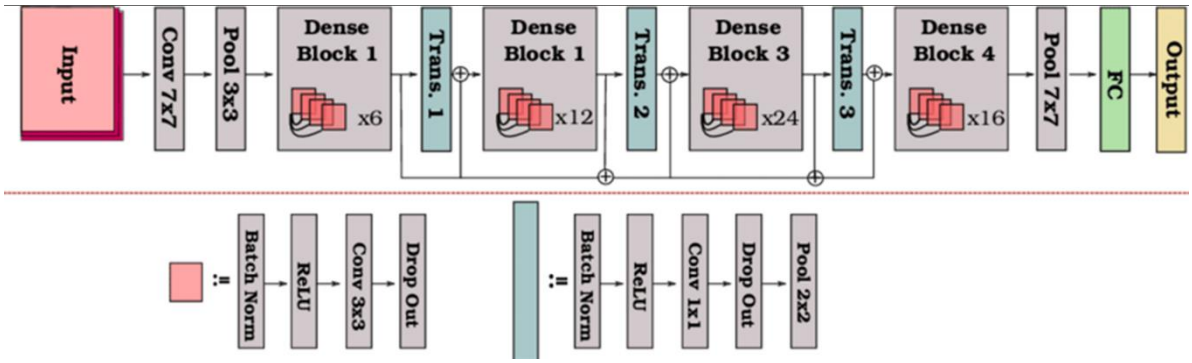


Figure 5.4

DenseNet-121 model was trained with the same dataset which was consist of 18,807 images in which 70% images were kept for training and 30% for testing. In this model 2 Dense layers were used with a sigmoid activation function in last layer, the model was executed with 20 epochs.

The training and validation accuracy is shown in figure 5.5

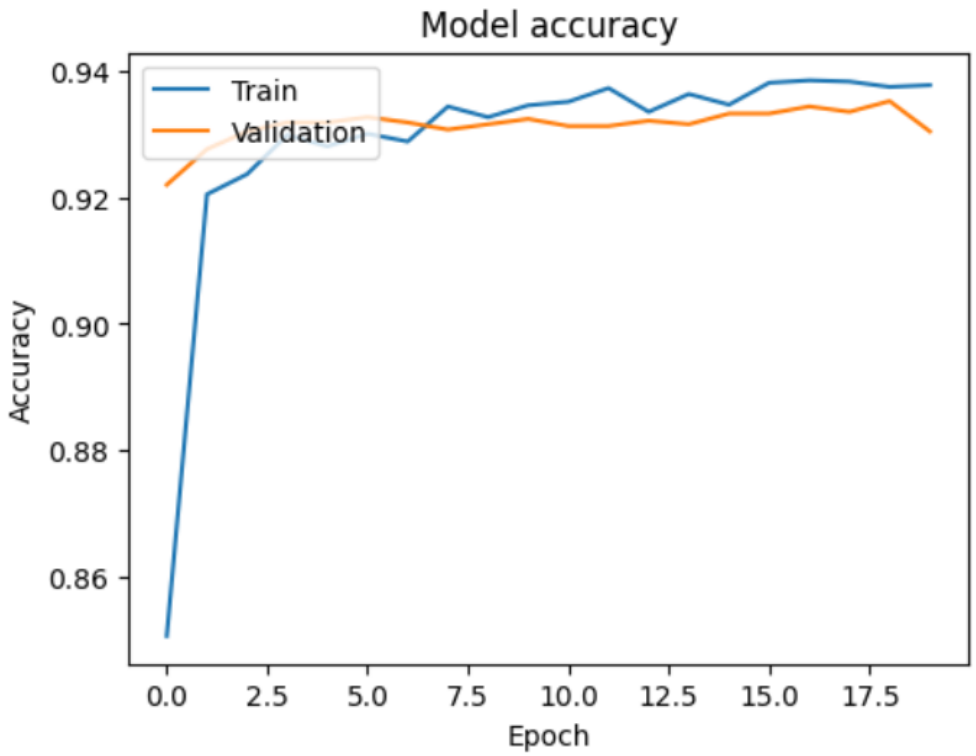


Figure 5.5

Figure 5.5 shows the training and validation accuracy of pre-trained model DesnseNet-121. The training accuracy in starting was 76 and the validation value was began from 92. In each epoch the training and validation accuracy was improving, at the last epoch the model achieved 93% of accuracy.

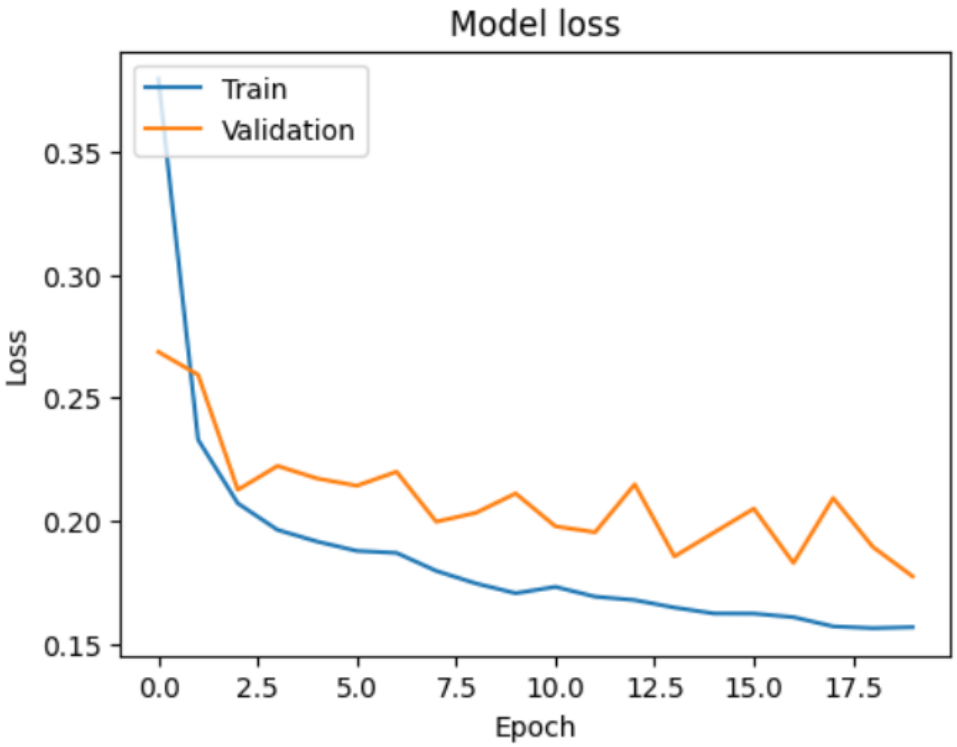


Figure 5.6

Figure 5.6 shows the DenseNet-121 pre-trained model training and validation loss, the model training loss in beginning was 49 and validation loss value was 26. During training of DenseNet-121 model, training and validation loss were continuously decreasing, in last epoch the training loss value was 15 and validation loss value was 17.

5.2.3 VGG-16

VGG-16 is another pre-trained model that was employed in this study. Using a series of convolutional layers, VGG-16—which is renowned for its simple yet effective design—first harvests features from the input images and then uses fully connected layers for classification with the sigmoid activation function in last layer. Using the same dataset, the model was trained. The VGG-16 model is widely recognized for its proficiency in image classification. The utilization of this model in this system provided significant insights into its melanoma detection capabilities. Architecture of VGG-16 is shown in Figure 5.7.



Figure 5.7

VGG-16 model was trained using the dataset of 18,807 images total, in this dataset 70% images were kept for training as well and 30% for testing. 20 epochs were used for training VGG-16 model.

Figure 5.8 shows training and validation accuracy of VGG-16 model.

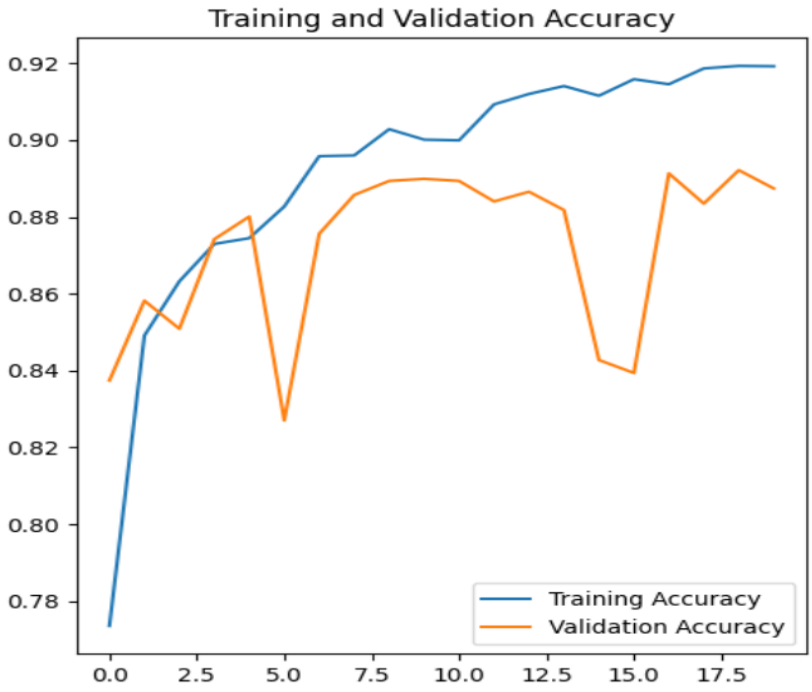


Figure 5.8

The training and validation accuracy of VGG-16 model is shown in figure 5.8. In first epoch the training accuracy starts from 69 and the validation accuracy from 83. Both training and validation accuracy was improving continuously till last epoch, till the last epoch the model achieved 88% accuracy.

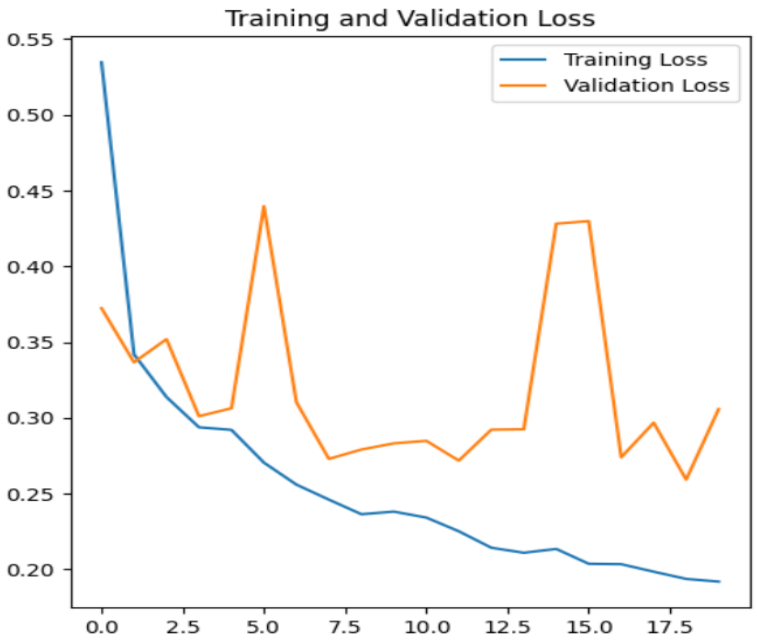


Figure 5.9

Figure 5.9 shows the training and validation loss of VGG-16 model. When Model start training in that time initially in first epoch the training accuracy loss was 87 and the loss was decreasing continuously till last epoch, in last epoch the training loss was 22. In this way the validation loss in start was 37 and also the validation loss was decreased to 29 when reached to last epoch.

5.2.4 MobileNetV2

MobileNetV2, which is designed to work effectively on mobile and edge devices, was also tested on this system. Its light-weight architecture and ability to achieve high accuracy level with fewer parameters make it perfect for resource-constrained applications. MobileNetV2 was designed with melanoma detection in mind and provided a quick and efficient solution for real-time detection tasks, even if its depth may not be as great as that of DenseNet-121 or VGG-16. This approach is very beneficial in scenarios where computational resources are limited, for as in mobile health applications. MobileNetV2 architecture is shown in figure 5.10.

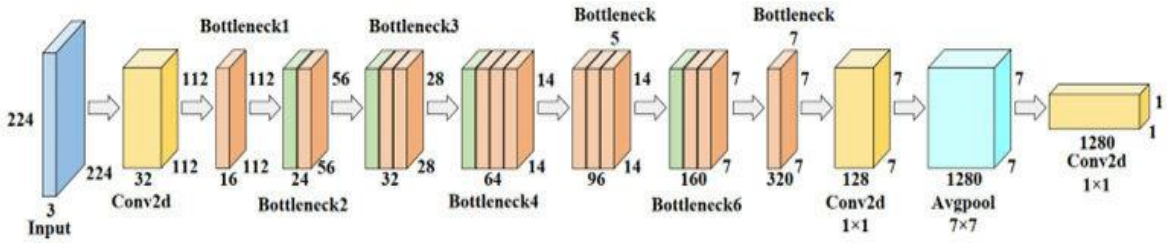


Figure 5.10

MobileNetV2 model as others trained with the same dataset with total of 18,807 images and 70% images kept for training and 30% for testing. For training of model 20 epochs were used.

After training the model a graph was plotted to show the model training and test accuracy and also for loss the loss graph was plotted which are shown in Figure 5.11 and Figure 5.12.

MobileNetV2 pre-trained model training and validation accuracy are shown in Figure 5.11

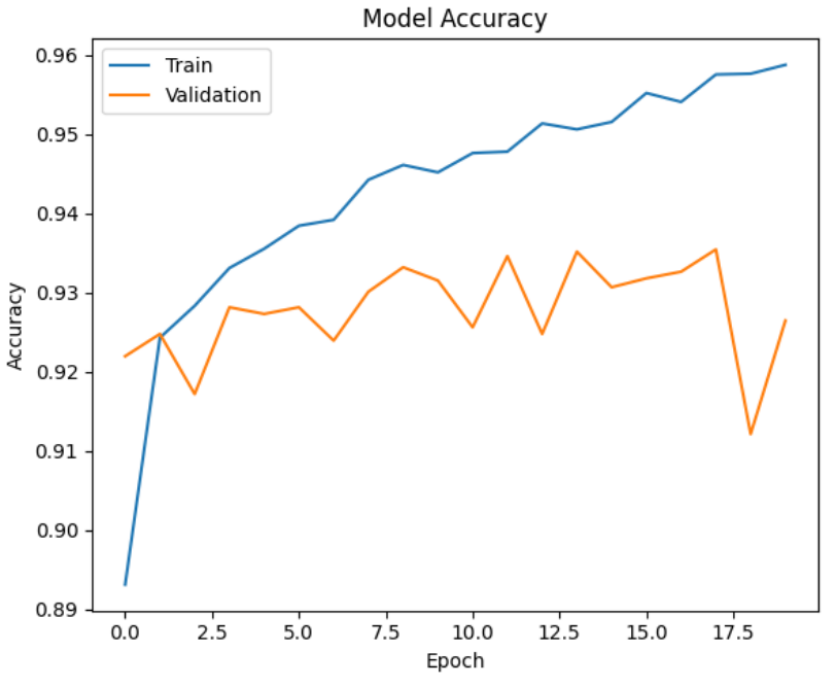


Figure 5.11

While training the MobileNetV2 model in first epoch the training accuracy was 83 and validation accuracy was 92 in start. And both accuracies were improving continuously but the validation accuracy in some epochs was down and up and in last epoch the model achieved 92% accuracy

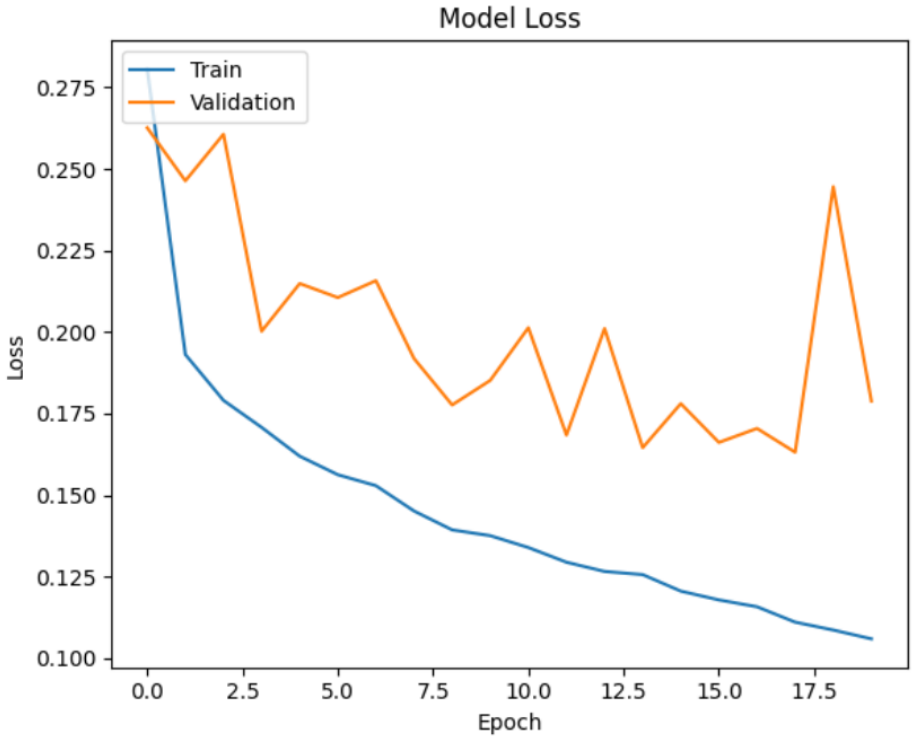


Figure 5.12

Figure 5.12 shows the MobileNetV2 pre-trained model training and validation loss, the model training loss in first epoch was 38 and validation loss value was 26. During training of MobileNetV2 model, training and validation loss were continuously decreasing, in last epoch the training loss value was 10 and validation loss value was 17.

5.3 Comparative Analysis of Models

The custom model, DenseNet-121, VGG-16, and MobileNetV2 capacity to recognize melanoma in real-world scenarios used as the benchmark for comparison. In terms of feature extraction and classification accuracy, two pre-trained models (DenseNet-121, MobileNetV2) performed better than the custom model, although the custom model provided a useful baseline. DenseNet-121 proved to be the most effective model due to its robust connections and ability to extract tiny details from pictures of skin lesions. Although VGG-16's efficiency was not as high as that of DenseNet-121, it was still rather good. For real-time detection in low-resource scenarios, MobileNetV2 and custom CNN models were the best option even though it had the maximum computational efficiency.

Table 5.1: Accuracy Comparison of trained models

Model	Accuracy %
CNN	91%
DenseNet-121	93%
VGG-16	88%
MobileNetV2	92%

Table 5.2: Loss Comparison of trained models

Model	Loss %
CNN	23%
DenseNet-121	17%
VGG-16	29%
MobileNetV2	17%

5.4 Limitations in the system

In time of testing and development of system some limitations were identified which are discussed below:

- **Resource Calculation:** In many computer resources, GPU is one of them needed to train the DL model like DenseNet-121. Performance will not so good if the system is installed on local machine that have not much powerful resources even if cloud services were utilizes for training.
- **Can't detect other skin cancer images than melanoma:** The main goal and purpose

of this system is detection of melanoma and it cannot used to detect other skin cancers.

- **Real-Time Scalability:** Scaling this system to handle larger number of user real-time requests may challenging.
- **Model Bias:** If we take steps to fix data imbalance, there is a possibility that model can still provide biased prediction. This problem can happen if the images used in prediction are different from the images through that model was trained on. To resolve this issue, the dataset updated with more disparate or relevant images to improve the model accuracy and performance.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Introduction

In this chapter, the outcomes of MDS are summarized and it reflects on overall success. The project which is completed, achieved the goal of building an accurate and efficient system that uses the DL models for real-time and accurate detection. In this chapter we discussed the essential achievements of this project, which include development of user-friendly interface, model selection, and backend integration. As well, the recommendations for future work are provided, where we highlighted the areas where the system can be improved in performance and usability more.

6.2 Summary and Conclusion

This project was setup for development of an intelligent system to detect MSC by using well performed and accurate DL models, with the goal to facilitate the early detection through a user-friendly interface. Due to its aggressive nature of disease melanoma detection is critical, treatment outcomes can be significantly improve because of early detection. Our focus was to develop a system that should provide accurate results and not only this, we also focused on creating a system which can be accessible to both general people (patients) and medical experts. All through this project, we trained and evaluate multiple model that are one custom CNN model and three pre-trained models (DenseNet-121, MobileNetV2, and VGG-16). After analysis and testing of all trained models, DesnseNet-121 was selected as the final model to be integrate into the system user interface because of high accuracy and good performance in prediction.

Through system user-interface users can upload images of skin lesions and receive prediction results quickly in seconds, providing a helpful tool to early detect MSC.

The project backend was built with the help of Django and integrated with MySQL database, that allows users to store their details such as patient id, name, age and also store their prediction results and also users can securely access their data.

The project frontend is designed to be user-friendly, interactive and allows users to use the system without any technical expertise easily.

It determines the demonstration of DL model potential of DL model and offer practical, scalable solution for detection of melanoma in early stages.

6.3 Achievements of the project

Following are the number of significant milepost accomplished by MDS:

i) **Integration of DL model:** we have trained 3 pre-trained models such as MobileNetV2, VGG-16, DenseNet-121, and one custom model. DenseNet-121 was selected as the final model because of its great accuracy and performance for integration.

ii) **Responsive User-friendly Interface:** Using Django we created a responsive web interface, and making it easier for users to interact with system to upload images and get the prediction results fast.

iii) **Backend and Database Integration:** For securely storing uploaded images, prediction results with the confidence value, patient details such as patient-id, name age and date, MySQL database was implemented for this purpose.

iv) **Real-time Melanoma Detection:** The system prediction process in real-time is so fast and quick, after uploading the skin lesion image user can get the result in seconds. This fast and quick response is so helpful and big achievement of this project in detection of melanoma in early stages for timely medical treatment.

v) **Accuracy:** After training different pre-trained models such as DenseNet-121, MobileNetV2, VGG-16, and one custom CNN model, DenseNet-121 was selected because of its great accuracy and high performance in detecting melanoma from skin images.

6.4 Recommendation for future work

Future work could focus on several key areas in which enhancement in MDS can be done. First, extending the dataset by adding more diverse skin types, ages and lesion can improve the generalizability and accuracy. By collaboration with medical institution for verified clinical datasets can further improve and boost the performance of the model. Furthermore, we can also add the functionality of detection for other types of skin cancer, and it can make the system more comprehensive. To develop a mobile app it can improve the accessibility for users to easily and quickly capture images with the mobile phone camera and perform detection by using mobile. We can also built the trust on model predictions by using a tool called Grad-CAM to explain and visually show which part of the image was important for model in prediction. At last, system can handle multiple users by deploying the system on scalable platforms like AWS or Google Cloud, to deliver reliable and fast results.

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